Chapter 4

Data-Driven Temporal Computing with Polymer-Electrolyte Memristor Reservoirs

4.1 Introduction and Scope

???? treated the slow, volatile response of these polymer-electrolyte composites first as a phenomenon to be demonstrated and then as a property to be controlled. This chapter treats it as a resource to be exploited. The guiding idea is simple: a device whose conductance carries a fading record of its recent input is, in miniature, a temporal processor, and a collection of such devices is a *reservoir* — a physical substrate that projects a time-varying input into a high-dimensional state from which simple, linear read-outs can solve temporal tasks.

The chapter rests on a single thesis that connects it to **??**: composition and drive *set* the device timescale, and a task *selects* the timescale it needs.

The design problem is therefore one of *matching*, and the open question is what a *spread* of timescales — the cheap, intrinsic heterogeneity of a solution-processed device family — buys on top of a single well-chosen one. The evidence is organised as three tiers of increasing realism. First, two task-agnostic reservoir benchmarks (memory capacity and NARMA-10) establish reservoir competence and show, cleanly, that timescale heterogeneity broadens recoverable memory. Second, a benchmark built from real physiological recordings asks the reservoir to reconstruct the multi-lag temporal context of affective signals, where heterogeneity again pays. Third, the downstream affective-classification task itself is reported honestly, including the finding that, once a single slow timescale dominates the discriminative information, heterogeneity is no longer required.

The evidence base is deliberately constrained. Every result is *in silico*: each device node is a compact behavioural model whose parameters are taken directly from the **??** fits (section 4.2), and every read-out is a single linear regression, so that no claimed computation can be smuggled in through a nonlinear decoder or a hand-tuned dynamical system. No new devices are fabricated. Two demonstrations run throughout: Demonstration A asks whether a *single* composition suffices for a single-timescale feature, and Demonstration B asks what a heterogeneous *bank* adds when the target is multi-timescale. The honest answer, assembled across the three tiers, is that the devices perform genuine temporal computing, that fading memory is the active ingredient, and that timescale heterogeneity is a measurable computational resource for memory breadth and a generality hedge — rather than a guaranteed improvement on every task.

4.2 From Measured Dynamics to a Behavioural Model

?? established that the fading memory of these composites is an engineerable quantity, dominated by composition. To put that memory to work, this chapter reduces each composition cell to a compact behavioural model and treats it as the unit cell of a reservoir computer. The model is deliberately minimal: it is fitted only to the three common measurements of **??** — hysteresis, pulsed potentiation, and delayed depotentiation — so that every simulation in this chapter traces back to measured data rather than to a hand-tuned dynamical system.

Each composition cell is summarised by a *parameter card*: the write nonlinearity ϕ_c from the potentiation curves, the fading-memory law λ_c from the depotentiation curves, and an associated device-to-device spread. The fading memory is the stretched-exponential (Kohlrausch) retention identified in **??**,

$$\lambda_c(\Delta t) \;=\; \exp\Big[-(\Delta t/\tau_c)^{\beta_c}\Big],$$

where τ_c and β_c are the per-cell relaxation time and stretch exponent; for the few cells without an identified stretched fit, a single exponential reproducing the model-free half-life is used instead ($\tau_c = t_{1/2}/\ln 2$, $\beta_c = 1$). The write nonlinearity is the compressive, saturating build-up of conductance with pulse number measured in **??**, parametrised by its log–log growth exponent α_c and its peak enhancement.

Driven by a rate-coded input $u_n \in [0, 1]$ sampled at interval Δt, a single device node updates as

$$x_n^{(i)} \;=\; \lambda_i\, x_{n-1}^{(i)} \;+\; \Big[\max(w_i\, u_n,\, 0)\Big]^{\alpha_i}, \qquad \lambda_i = \exp\Big[-(\Delta t/\tau_i)^{\beta_i}\Big],$$

so that the node integrates a compressive write of its (masked) input and then leaks by exactly the measured retention λ_i over the sample interval;

w_i is the reservoir input mask. The internal state is read sub-threshold and is therefore assumed state-preserving, as in the proof-of-concept device of **??**. Every read-out in this chapter is a single *linear* ridge regression on the device states, so that any computation demonstrated is attributable to the devices and not to a nonlinear read-out.

Two honesty constraints are carried explicitly from **??** into every result below. First, the write nonlinearity and the fading-memory law were measured at *different write amplitudes* (3 V for potentiation, 4 V for depotentiation; **??**); composing them in a single node is an explicit modelling assumption, not a measured fact, and it most likely underestimates the true coupling between drive and retention. Second, both were measured at a single, effectively common inter-pulse cadence, so the model is not entitled to claims about input timing finer than that cadence. The matched-amplitude, varied-cadence pulse-train experiment that would remove both caveats is identified as the key future bench test (section 4.9).

4.3 Reservoir Computing and the Common Metric Language

The reservoir principle. Reservoir computing emerged from the observation that a fixed, randomly-parametrised dynamical system — the *reservoir* — can perform useful temporal computation provided only that a *linear* read-out is trained on its state, leaving the dynamics themselves untrained [1–3]. Two properties make a physical system a competent reservoir: a *fading memory* of past inputs, so that the state reflects recent history but eventually forgets, and a *nonlinear* response, so that the state mixes inputs in ways a linear read-out can exploit. These are precisely the two properties characterised in **??** — the stretched-exponential depotentiation supplies the fading memory, the compressive potentiation supplies the nonlinearity — which is what makes a bank of these devices a natural physical reservoir. The same recognition has driven a wave of memristive and organic reservoir demonstrations [4–7], of which the polymer-electrolyte devices studied here are a low-power, biocompatible instance.

Architecture. The reservoir used throughout this chapter is a spatial bank of N *independent* device nodes, each a single behavioural model (section 4.2) driven by a randomly-masked copy of the input. Independence is a deliberately conservative choice: with no inter-node recurrence, all of the reservoir's richness must come from the spread of intrinsic timescales across compositions, the device-to-device scatter, and the input mask, so any benefit attributed to heterogeneity cannot have been inherited from engineered connectivity. Where a single composition is the subject (Demonstration A), the same node can equally be deployed as a time-multiplexed, delay-feedback reservoir whose virtual nodes are distributed along a delay line [8]; the two architectures are interchangeable for the present claims, and the spatial bank is used for uniformity.

One ruler. So that the demonstrations remain comparable, every result is reported on a common set of metrics. The linear *memory capacity* MC_k and its total [9] quantify how much past input is linearly recoverable at each lag, and so visualise timescale coverage directly; NARMA-10 [10] is a standard nonlinear system-identification benchmark; the information-processing-capacity framework [11] underlies the separation of linear from nonlinear contributions; and the affective tasks are scored by class-balanced macro-F1. Robustness is probed by injecting the measured device-to-device and cycle-to-cycle spread into the bank.

4.4 Benchmark Tier: Memory Capacity and NARMA

The first tier of evidence is deliberately task-agnostic: two standard reservoir-computing benchmarks that establish reservoir capability and isolate the contribution of timescale heterogeneity without reference to any application. Both use a random rate-coded input, a bank of N device nodes, and a linear ridge read-out, and both are reported on the same metric ruler so that the two demonstrations of this chapter remain directly comparable.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can linearly reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total memory capacity is the sum of MC_k over all lags [9]. Figure 4.1 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09, differing only by device-to-device scatter) and a *heterogeneous* bank spanning the composition grid (τ from ≈ 3 to $\approx 26\,\mathrm{s}$). The heterogeneous bank broadens the memory profile across lags and raises the total memory capacity from 4.10 to 6.12, a factor of 1.49, with near-perfect single-step recall (MC_1 rising from 0.56 to 1.00) and the gain concentrated at short-to-intermediate lags. Because the nodes are independent — a bank with no inter-node recurrence — this is deliberately the hard case for a reservoir, and the heterogeneity benefit, which arises purely from the spread of timescales and the device scatter, is therefore a conservative estimate.

Composition sweep and the single-node optimum. The same benchmarks, run on single-composition banks, rank the compositions

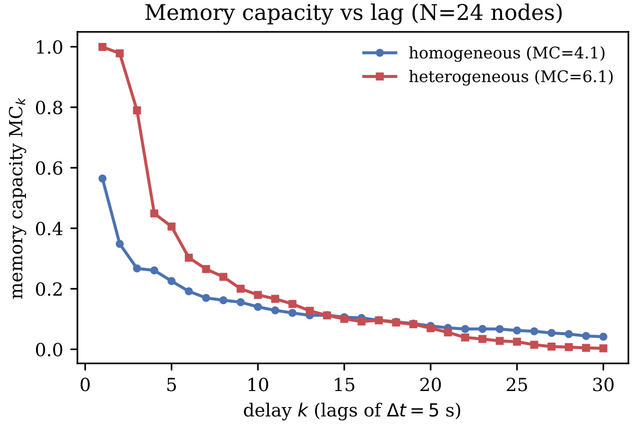

Figure 4.1: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 3$–$26\,$s). The heterogeneous bank broadens recoverable memory across lags, raising total memory capacity from 4.10 to 6.12 (a 1.49× increase), with the gain concentrated at short-to-intermediate lags. $N = 24$ leaky nodes, random uniform input, linear (ridge) read-out; in-silico devices built from the Chapter 3 parameter cards.

against one another and so test the Demonstration-A claim that one cell suffices for a single-timescale task. Figure 4.2 shows that the lead cell PEO 0.3/0.09 attains the highest single-composition memory capacity (3.44), while the neighbouring PEO 0.3/0.045 cell is marginally better on the NARMA-10 system-identification benchmark [10] (NRMSE 0.627 against the lead's 0.661, a close second). Both optima fall in the low-PEO row that **??** identifies as carrying the longer fading-memory times, so the benchmark ranking is consistent with the composition result rather than an artefact of the simulation. This earns the Demonstration-A statement: for a single-timescale feature, the lead composition is the memory-capacity optimum and a heterogeneous bank is unnecessary. Only the *relative* ranking is claimed; the absolute NARMA-10 error is high (≈ 0.6–0.8) precisely because an independent-node bank is a weak NARMA-10 reservoir, and this is stated rather than hidden.

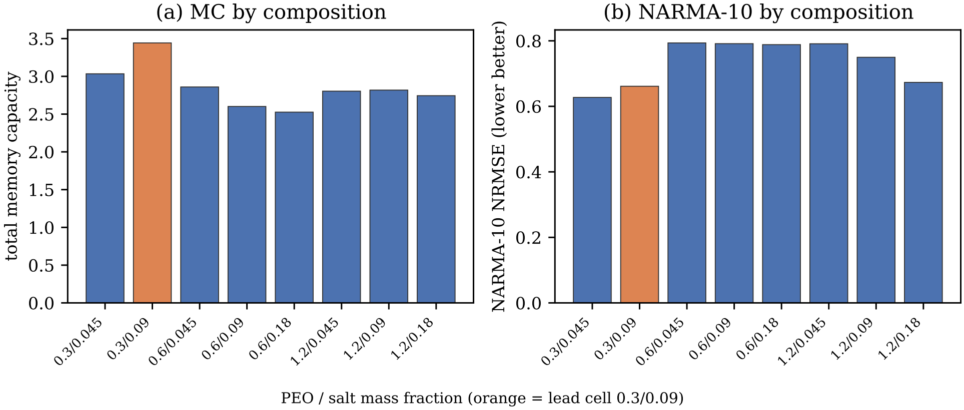

Figure 4.2: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The lead cell PEO 0.3/0.09 (orange) attains the highest memory capacity; the neighbouring 0.3/0.045 cell is marginally better on NARMA-10. Both optima fall in the low-PEO row anticipated by **??**. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

4.5 Affective Computing: A Native Timescale Match

Affective computing — the recognition of human emotional and physiological state by machines — was framed by Picard as a route to more natural human–computer interaction, classically from *peripheral* physiological signals that are slow, wearable-friendly, and unobtrusive to acquire [12]. It is the flagship domain of this chapter, for two reasons that follow directly from **??**. First, the device timescale window — fading-memory times from a few seconds to tens of seconds — sits natively on top of the bands that carry affective information, so no aggressive time-rescaling is needed to bring signal and substrate into register (table 4.1). Second, affective information is *intrinsically multi-timescale*: a fast phasic electrodermal response, a respiration- and heart-rate-variability rhythm of a few seconds,

Table 4.1: Characteristic timescales of the peripheral affective signals used in this chapter, and the composition cells whose fading-memory window covers them [13–15]. The grid of **??** spans the full range, with the lead PEO 0.3 cells serving the slow LF-HRV band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	$\sim$1–10 s	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	$\sim$3–5 s	mid PEO
HRV, HF band	$\sim$2.5–7 s	mid PEO
HRV, LF band	$\sim$7–25 s	PEO 0.3 ($\tau \approx$ 19–26 s)
Tonic SCL	minutes	slowest / drive-boosted

and a tonic level that drifts over minutes all carry state at once [13–15]. That structure is the honest, signal-driven motivation for a heterogeneous bank, and it is also the deployment narrative for an organic, flexible, biocompatible, low-power substrate worn on the skin.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [16]: fifteen subjects wearing a chest sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways — as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 4.6), and as a downstream affective-classification task (section 4.7).

4.6 Physiological Temporal-Context Reconstruction

The benchmarks of section 4.4 prove that a spread of device timescales broadens memory on a *random* input. This section puts that property to a more demanding test on the real, intrinsically multi-timescale affective signals introduced in section 4.5, isolating the multi-timescale *encoding* question before turning to final labels (section 4.7).

The reservoir is run continuously over each WESAD session at $\Delta t = 1\,\mathrm{s}$ on four chest channels — electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate — and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [13, 14]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Table 4.2 reports the held-out reconstruction quality. A memoryless bank barely improves on instantaneous input ($R^2 = 0.674$ against 0.673): with no fading memory, the past is simply not available. Memory of a single timescale lifts R^2 to 0.74, and the heterogeneous composition bank reaches the best single-bank score, 0.756 — a gain of +0.084 over instantaneous input and +0.012 over the best same-size homogeneous control. The mechanism is visible in the band-resolved columns: the fast homogeneous bank is best at the fastest context and the slow homogeneous bank at the slowest, but neither covers the whole vector, whereas the heterogeneous bank is the best *single* bank across the full fast-to-slow physiological context (fig. 4.3). This is the affect-aligned statement of "heterogeneity is a computational resource" on real signals, and it is exactly the result that the final-label classifier of section 4.7 cannot expose.

4.7 WESAD Affective Classification

The final tier is the downstream task itself: classifying affective state from the WESAD physiological streams with the device reservoir and a linear read-out, leave-one-subject-out. It is reported here as an honest

Table 4.2: Leave-one-subject-out held-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. The heterogeneous bank gives the best single-bank score across the full context vector; each homogeneous bank wins only its own band.

Bank	overall R^2	fast	mid	slow
Instantaneous	0.673	0.715	0.670	0.632
Memoryless	0.674	0.721	0.667	0.630
Homogeneous, fast	0.741	**0.815**	0.735	0.669
Homogeneous, slow	0.744	0.787	0.724	**0.711**
Heterogeneous	**0.756**	**0.815**	**0.737**	0.708

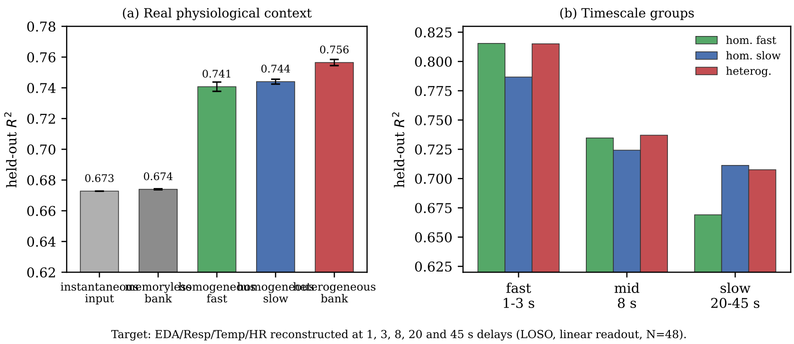

Figure 4.3: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and ECG-derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s from the reservoir state (leave-one-subject-out, $N = 48$). The heterogeneous composition bank gives the best single-bank held-out R^2 (0.756) across the full fast-to-slow context vector, ahead of the best same-size homogeneous control (0.744) and well ahead of instantaneous input (0.673). Fast and slow homogeneous banks each win only their own band.

application result — the devices perform real affective computing, and the contribution of timescale heterogeneity to final labels is reported with error bars, including where it is null.

Demonstration A: a single composition suffices for a single-timescale feature. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches a leave-one-subject-out macro-F1 of 0.894 (fig. 4.4a). This confirms the Demonstration-A claim that one well-matched composition is a competent affect node. It comes, however, with an instructive caveat: a static feature baseline (the mean, standard deviation, last value and slope of the same signal) matches it (0.899). The 60-second-window task is therefore quasi-static — the discriminative information sits in the instantaneous signal level, not in its temporal history — which is exactly why timescale heterogeneity has nothing to contribute here, and why a memory-demanding reformulation is needed to test the heterogeneity claim fairly.

Demonstration B: streaming three-class tracking. The fair test is a streaming one: the reservoir runs continuously over each session and the affective class (baseline, stress, or amusement) is read out at *every* step, so that performance genuinely depends on fading memory. Table 4.3 decomposes the result into four matched-size banks. Relative to instantaneous input, adding dimensionality (a same-size memoryless bank) contributes +0.028, adding fading memory of a single timescale a further +0.016, and adding timescale heterogeneity a final +0.005. The genuine fading-memory gain over a same-size memoryless bank is a modest but consistent +0.014 to +0.020. The timescale-heterogeneity increment, however, is $+0.005 \pm 0.011$ — within seed noise (positive in seven of ten seeds; across sampling intervals from 0.5 to 4 s the heterogeneous-minus-homogeneous gap stays in $[+0.002, +0.007]$ but never separates from zero). Per class, the best bank reaches F1 0.90 on baseline and 0.92 on stress but only 0.55 on the short, low-arousal amusement class, which is the hard class for every model.

Table 4.3: Leave-one-subject-out macro-F1 for streaming baseline/stress/amusement tracking on WESAD (means ± SD over ten bank seeds; matched input masks across banks). Each increment is the gain over the row above: fading memory contributes a small, consistent gain, while the timescale-heterogeneity increment is within noise (*ns*).

Bank	macro-F1	increment
Instantaneous (no memory)	0.709	—
Memoryless (dimensionality only)	0.737	+0.028
Homogeneous (memory, single τ)	0.753 ± 0.009	+0.016
Heterogeneous (memory, τ spread)	0.758 ± 0.006	+0.005 (*ns*)

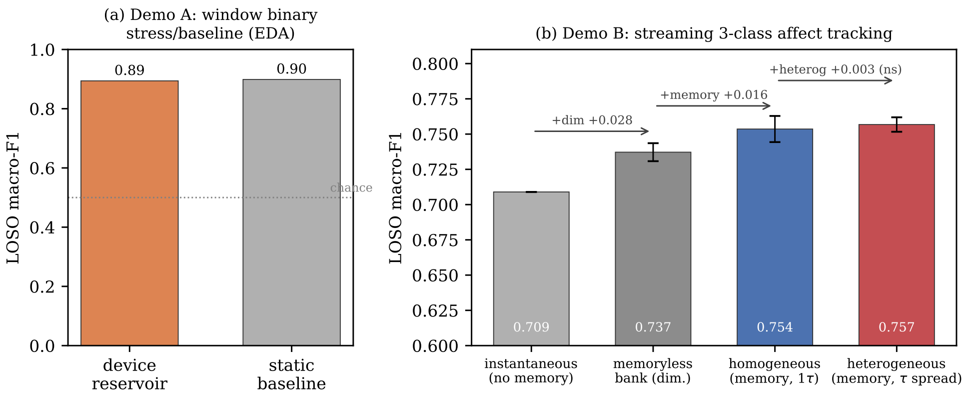

Figure 4.4: Affective computing with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Demonstration A, window-level binary stress-versus-baseline from EDA: the single lead composition matches a static feature baseline, i.e. the 60 s-window task is quasi-static and requires no fading memory. (b) Demonstration B, streaming three-class affect tracking (baseline/stress/amusement) decomposed as instantaneous → +dimensionality → +fading memory → +timescale heterogeneity. Fading memory contributes a small but consistent gain; the timescale-heterogeneity increment ($+0.005 \pm 0.011$) is within seed noise. Error bars are ±1 SD over 10 bank seeds.

Why heterogeneity does not pay off on the labels. The null is not a failure of the devices but a property of the task. Affect labels are slowly varying, sustained states; their discriminative memory demand is dominated by a single slow, tonic band that the lead composition's relaxation time ($\tau \approx 19$–$26\,\mathrm{s}$) already serves. Timescale heterogeneity converts into task performance only when the target requires memory at many distinct lags *simultaneously* — which the memory-capacity benchmark (fig. 4.1) and the physiological-context reconstruction (table 4.2) both demand, and on which heterogeneity does pay, but which a single sustained label set does not. The honest reading, carried into section 4.8, is that the devices do real affective computing, that fading memory helps, and that composition-timescale *matching* is the operative design lever — with heterogeneity a generality and robustness hedge rather than a guaranteed win on any one label set.

4.8 Judging the Chapter 3 Findings Against the Application

Sections 4.4, 4.6 and 4.7 together close the loop opened in **??**. That chapter showed that composition — and, as a methodological corollary, drive amplitude — *sets* the fading-memory timescale of these devices. This chapter shows the other half of the design rule: an application *selects* a timescale, or a set of timescales, and performance follows from how well the device family covers them. The composition sweep makes the rule concrete in one direction (fig. 4.2): the lead cell is the memory-capacity optimum because its relaxation time best matches a single seconds-scale target, which is the regime of Demonstration A. The multi-lag results make it concrete in the other (table 4.2): when the target spans bands that no single cell covers, the bank whose timescales are *spread* to match them is the best single choice.

Heterogeneity as a design resource. The three tiers give a consistent — and honest — account of what timescale heterogeneity is worth. Where the task demands memory at many lags at once, namely random-input memory capacity (a 1.49× broadening; fig. 4.1) and real-physiology context reconstruction (+0.012 in held-out R^2 over the best homogeneous bank; table 4.2), heterogeneity is a measurable computational resource. Where a single slow timescale already dominates the discriminative information, as in the sustained WESAD affect labels, it is a generality and robustness hedge rather than a source of additional accuracy. Crucially for the materials argument of this thesis, that heterogeneity is *cheap and intrinsic*: it comes for free with a solution-processed device family, through composition spread, device-to-device scatter, and — as further, illustrative knobs catalogued in **??** but not required by the quantitative banks here — electrolyte chemistry and drive amplitude.

Operating point and integration. The same parameter cards that drive the simulations also fix the deployment envelope. The devices operate in the sub-kilohertz regime in which affective physiology lives, so the slow response that disqualifies them for fast logic is exactly what suits them to on-skin sensing. The read-out is assumed sub-threshold and state-preserving, which sets a read-voltage and read-disturb budget that any circuit implementation must respect; and the device-to-device scatter that a memory application would treat as a yield problem is, in the reservoir setting, part of the computational substrate. These are the design rules that **??** makes available and that an eventual hardware reservoir would inherit.

4.9 Limitations and Scope Conditions

The claims of this chapter are bounded by scope conditions that are stated here in full, in keeping with the claim discipline of **??**.

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 4.2, not a bench reservoir, and every figure caption names the data behind it; the step from a faithful per-device model to a wired hardware reservoir is left to future work.

The composition assumption. As detailed in section 4.2, the write nonlinearity and the fading-memory law were measured separately, at different write amplitudes and a single inter-pulse cadence. Composing them assumes the two regimes are compatible and probably under-estimates the coupling between drive and retention. A single matched-amplitude experiment that sweeps pulse count and inter-pulse interval together would remove the assumption; it is the most important bench experiment this chapter calls for.

Operating regime and read-out. The model is identified, and valid, only in the slow, sub-kilohertz regime of the measurements; affective signals fall natively in this band, but fast temporal tasks are out of scope. Every read-out is a single ridge regression — a deliberate constraint, since a nonlinear read-out could solve the tasks by itself and would invalidate the reservoir claim — and the read is assumed sub-threshold and state-preserving.

Heterogeneity axes. The quantitative heterogeneous banks span the replicated PEO/LiOTf composition grid. The chemistry and drive-amplitude axes of **??**, being sample-limited ($n \leq 2$ per matched cell), enter only as illustrative additional knobs; Demonstration B is therefore a principle demonstration of composition heterogeneity, not a powered comparison across chemistries. The composition cells that define these banks index *composition*, not film thickness: although spin-coating speed and residual film thickness covary with the PEO fraction, the parameter cards inherited here were shown to be governed by composition rather than thickness (**??**, **??**), so the heterogeneity exploited in this chapter is compositional in origin.

A single dataset, and an honest null. The affective results rest on one corpus (WESAD). The central negative result — that timescale heterogeneity does not separably improve final-label classification — is reported with error bars across seeds and sampling intervals rather than omitted, because it is informative: it localises exactly where heterogeneity helps and where it does not.

4.10 Conclusions and Bridge to Chapter 5

This chapter has shown that a bank of solution-processed polymer-electrolyte memristors, characterised through nothing more than the three common measurements of **??**, performs genuine temporal computing. Reduced to compact, measured behavioural models and read out by a single linear regression, the devices reach competitive scores on the standard memory-capacity and NARMA-10 reservoir benchmarks, encode the multi-lag temporal context of real physiological signals, and classify affective state from the WESAD corpus at a leave-one-subject-out macro-F1 of 0.894 on a binary stress task and 0.758 on a three-class streaming task.

The results resolve into a single design rule that unifies the two experimental chapters. **??** establishes that composition and drive set the device timescale; this chapter establishes that the application selects the timescale, that fading memory is the active computational ingredient, and that a spread of timescales is a measurable resource for memory breadth — a 1.49× broadening of memory capacity and a consistent gain in physiological-context encoding — even though it is not required when a single slow timescale already dominates a task. The heterogeneity that **??** catalogued as a tuning landscape is, in this light, not an imperfection to be suppressed but the raw material of the computation.

The concluding chapter draws together the contributions of the thesis and its limitations. The most consequential next steps follow directly from the scope conditions above: the matched-amplitude, varied-cadence pulse-train experiment that would convert the composition assumption into measured fact; a powered study of the chemistry and drive-amplitude heterogeneity axes; and a hardware reservoir that replaces the in-silico bank with wired devices, closing the gap between a faithful device model and a physical temporal computer.

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